

PSW Register

2bit $\rightarrow$ User Defined

6bit

 $\rightarrow$ 2bit bank $\rightarrow$ 4bit control

Until example 2.4

Program Status Word

2bit = 4 bank

PSW.4 PSW.3

00	0
01	1
10	2
11	3

Register Banks & Their Ram Addresses

B 0 $\rightarrow$ R0-R7 (0-7) 0, 1, 2, 3, 4, 5, 6, 7B 1 $\rightarrow$ R0-R7 (8-F) 8, 9, A, B, C, D, E, FB 2 $\rightarrow$ R0-R7 (10-17) 10, 11, 12, 13, 14, 15, 16, 17B 3 $\rightarrow$ R0-R7 (18-1F) 18, 19, 1A, 1B, 1C, 1D, 1E, 1F

Total bank bit = 2

Total bank addresses = 4

Total registers in bank(s) = $4 \times 8 = 32$ $\nearrow$ Example 2.5

MOV R0, #99H

MOV R1, #85H

 $\nearrow$ Example 2.7

SETB PSW 4

; select bank 2

MOV R0, #99H

; RAM location 10H has 99H

MOV R1, #85H

; RAM location 11H has 85H

Shayan

Stack

SP $\Rightarrow$ 07 when 8051 is powered up

Data adds from 08

PUSH = $\text{SP}++$ POP = $\text{SP}--$

Question : * PSW, 4

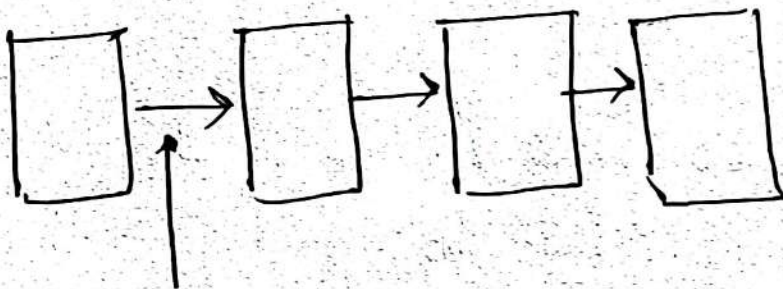
* Stack (Example 2.8)

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Address ধরে push করতে হয়

X R3

✓ "3" from 'bank 0' (by default)



use arrows

while answering

Example 2.9

17) haryan

3

Stack: Push & Pop

MOV SP, #5FH

MOV R1, #25H

MOV R2, #30H

MOV R4, #35H

MOV R5, #40H

PUSH 2

PUSH 4

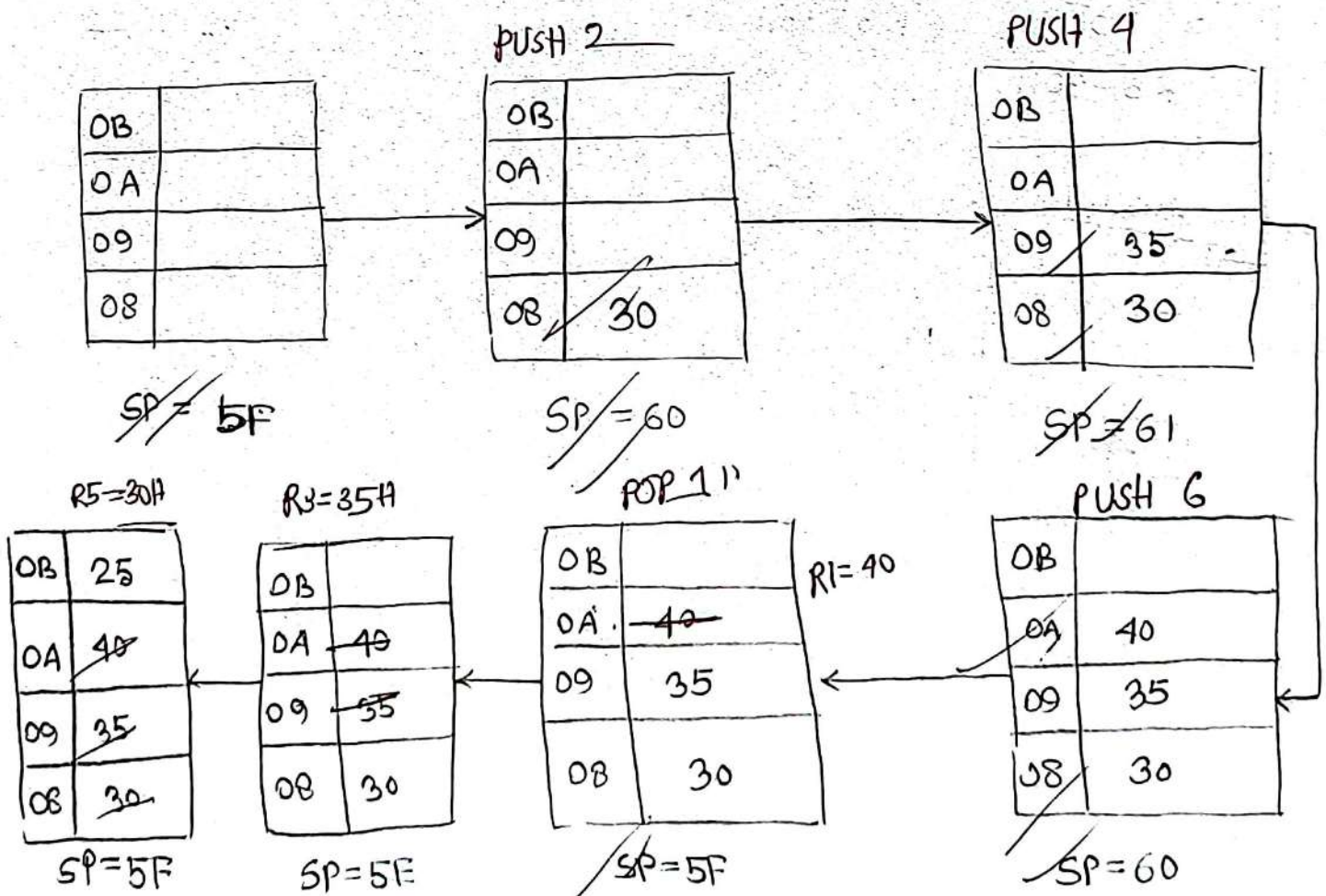
PUSH 6

POP 1

POP 3

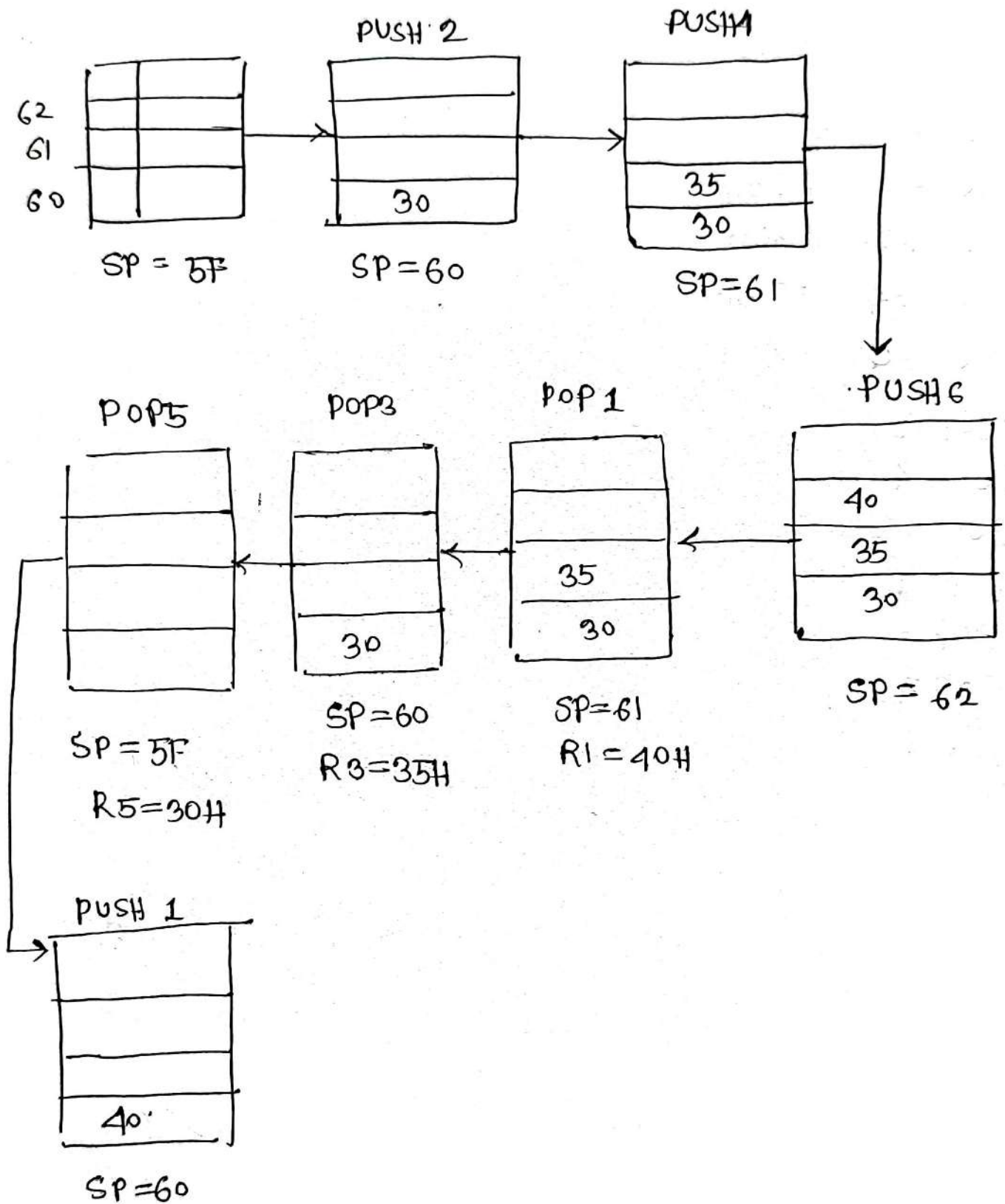
POP 5

PUSH 1



Shayan

1



rayan

→ Call instruction of Stack

* why SP needs to be incremented?

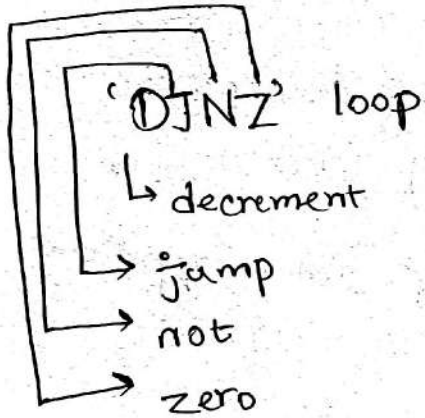
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* 8051 power up ~~रखल~~ register bank...

→ Example 2.10

'Chapter - IV'

JUMP, LOOP, & CALL Instruction



DJNZ reg, Label

(max loops 255 if value is FFH)

→ Program - 01

← A = 0 accumulator

Max loop = 10

So, reg = 10

MOV R2, #10H

label: A = 3 + A → 1st step

A = 3

A = 3 + A = 6

A = 3 + A = 9

i
10
9
8
⋮
0

→ Nested Loop ✓
→ page 106 example

Single loop দিয়ে
হবে না

$$\begin{aligned} 8\text{bit} &= 256 \\ &= 0-255 \end{aligned}$$

5 5
↙ ↘
0 1 0 1 0 1 0 1
1 0 1 0 1 0 1 0

A A

A = 55 → AA
AA → 55

✓ 106 page done

R3 = 10	R3 = 9
R2 = 70	R2 = 70
= 69	= 69
= 68	= 68
⋮	⋮
0	0

Outer loop শেষ হলে
Inner loop শেষ

Outer loop, inner loop কে
নিয়ন্ত্রিত করতে পারবে